

Chapter 15

Global Equity Risk Modeling

Jose Menchero, Andrei Morozov, and Peter Shepard

15.1 Introduction

The pioneering work of [Markowitz \(1952\)](#) formally established the intrinsic tradeoff between risk and return. This paradigm provided the foundation upon which the modern theory of finance was built and has proven so resilient that it has survived essentially intact for over 50 years. Almost as remarkable is the vigor with which the theory has been embraced by academics and practitioners alike.

The specific problem addressed by Markowitz was how to construct an efficient portfolio from a collection of risky assets. Markowitz defined an efficient portfolio as one that had the highest expected return for given level of risk, which he measured as standard deviation of portfolio returns. Markowitz showed that the relevant risk of an asset is not its stand-alone volatility, but rather its contribution to *portfolio* risk. Henceforth, the concepts of risk and correlation became inseparable.

A plot of expected return versus volatility for the set of all efficient portfolios maps out a curve known as the efficient frontier. In order to construct the efficient frontier using the Markowitz prescription, an investor must provide expected returns and covariances for the universe of all investable assets. The Markowitz procedure identifies the optimal portfolio corresponding to the risk tolerance of any given investor.

[Tobin \(1958\)](#) took the Markowitz methodology and extended it in a very simple way that nonetheless had profound implications for portfolio management. By including cash in the universe of investable assets, Tobin showed that there existed a *single* portfolio on the efficient frontier that, when combined with cash, dominated all other portfolios. For any investor, therefore, the optimal portfolio would always consist of a combination of cash and the “super-efficient” portfolio. For instance, risk-averse investors may combine the super-efficient portfolio with a large cash position, whereas risk seekers would borrow cash to purchase more of the super-efficient portfolio. Therefore, according to Tobin, the optimal investment strategy

J. Menchero (✉), A. Morozov, and P. Shepard
MSCI Barra 2100 Milvia Street, Berkley, CA 94704, USA
e-mail: jose.menchero@mscibarra.com

consists of two separate steps. The first is to determine the super-efficient portfolio. The second step is to determine the appropriate level of cash that matches the overall risk tolerance of the investor. This two-step investment process came to be known as the Tobin separation theorem.

The next major step in the development of Capital Market Theory was due to Sharpe (1964). By making certain assumptions, such as that all investors followed mean-variance preferences and agreed on the expected returns and covariances of all assets, he was able to show that the super-efficient portfolio was the market portfolio itself. Sharpe's theory, known as the Capital Asset Pricing Model, predicts that the expected return of an asset depends only on the expected return of the market and the beta of the asset relative to the market. In other words, within CAPM, the only "priced" factor is the market factor.

Using the CAPM framework, the return of any asset can be decomposed into a systematic component that is perfectly correlated with the market, and a residual component that is uncorrelated with the market. The CAPM predicts that the expected value of the residual return is zero. This does not preclude the possibility, however, of correlations among the residual returns. That is, even under the CAPM, there may be multiple sources of equity return co-movement, even if there is only one source of expected return.

Rosenberg (1974) was the first to develop multi-factor risk models to estimate the asset covariance matrix. This work was later extended by Rosenberg and Marathe (1975), who conducted a sweeping econometric analysis of multi-factor models. The intuition behind these models is that there exists a relatively parsimonious set of pervasive factors that drive asset returns. For instance, equity returns may be explained by country and industry membership, as well as by the exposure to style factors such as Size or Momentum. Returns that cannot be explained by the factors are deemed "stock specific" and are assumed to be uncorrelated. Rosenberg founded a firm, Barra, which made widespread the use of multi-factor risk models and dedicated itself to helping practitioners implement the theoretical insights of Markowitz, Tobin, Sharpe, and others. The first multi-factor risk model for the US market, dubbed USE1, was released in 1975.

The availability of rigorous multi-factor risk models such as USE1 enabled practitioners for the first time to apply the Markowitz procedure under realistic covariance assumptions. Even then, in the early days of portfolio optimization, practitioners were hampered by limitations in computational power. Consequently, much of the early research focused on the development of clever algorithms designed to overcome these obstacles, as described, for example, by Rudd and Rosenberg (1979).

Another major milestone on the path to practical implementation of modern portfolio theory came in 1989 with the development of the first Barra global equity risk model, dubbed GEM. This model was estimated via monthly cross-sectional regressions using countries, industries, and styles as explanatory factors, as described by Grinold et al. (1989).

In this paper, we present the latest Barra global equity risk model, GEM2. This model incorporates several advances and innovations over previous Barra global equity risk models. For instance, GEM2 employs an updated industry fac-

tor structure based on the *Global Industry Classification Standard* (GICS[®]), thus better reflecting the global economy's current industrial structure. The list of style factors has been expanded to capture additional sources of equity return co-movement. The global factor structure has been placed on an intuitive foundation with the introduction of a World factor, thereby treating industries and countries on an equal footing. GEM2 also utilizes higher frequency observations to allow for more accurate and timely risk forecasts. The GEM2 specific risk model is based on a structural approach that incorporates the latest advances in Barra risk model methodology. Finally, GEM2 uses a very broad estimation universe based on the MSCI *Global Investable Market Indices* (GIMI) for greater precision in factor return estimates.

GEM2 is available in short-term (GEM2S) and long-term (GEM2L) versions. Both versions share the same factor structure and have identical factor returns. They differ in their responsiveness. GEM2S is designed to provide users with the most accurate and responsive risk forecasts at the 1-month prediction horizon. GEM2L, by contrast, is intended for longer-term investors who value more stable risk forecasts. This paper describes only the short-term version, GEM2S. For notational efficiency, however, in this paper, we refer to this model as simply the GEM2 model.

The remainder of this paper is organized as follows. In Section 15.2, we provide an overview of the basic calculations involved in global equity risk forecasting. Section 15.3 describes the GEM2 factor structure. Section 15.4 describes estimation of the GEM2 factor returns. Section 15.5 discusses construction of the factor covariance matrix. Section 15.6 presents the specific risk model, followed by an evaluation of the accuracy of the risk forecasts in Section 15.7. This paper also contains two technical appendices. Appendix A provides details of the individual descriptors that comprise the style factors. Appendix B gives a review of bias statistics.

15.2 Forecasting Global Equity Portfolio Risk

Global investors derive return from two basic sources: price appreciation in the local currency of the asset, and repatriation of the asset value back to the base currency (numeraire) of the investor.

For concreteness, consider an investment from the perspective of a US portfolio manager. Let $C(n)$ denote the local currency of stock n , and let $r_{FX}^{C(n),\$}$ be the return of currency $C(n)$ due to exchange-rate fluctuations against the dollar. The return of stock n in US dollars, denoted $\hat{R}_n^\$$, is given by the usual expression

$$\hat{R}_n^\$ = \hat{r}_n + r_{FX}^{C(n),\$} + \hat{r}_n r_{FX}^{C(n),\$}, \quad (15.1)$$

where $\hat{r}_n$ is the local return of the asset. The cross term, $\hat{r}_n r_{FX}^{C(n),\$}$, is typically minute and can be safely ignored for risk purposes.¹

¹ A notable exception to this is during times of extreme inflation, when the cross term can become significant.

Our primary interest is to forecast the volatility of *excess* returns (i.e., above the risk-free rate). If $r_f^{\$}$ denotes the risk-free rate of US dollars, then $R_n^{\$} = \hat{R}_n^{\$} - r_f^{\$}$ is the excess return of stock n measured in US dollars.

Suppressing the cross term, and utilizing (15.1), we decompose the excess return into an equity component r_n and a currency component $q_n^{\$}$:

$$R_n^{\$} = r_n + q_n^{\$}. \quad (15.2)$$

The equity component is the *local excess return* of the stock,

$$r_n = \hat{r}_n - r_f^{C(n)}, \quad (15.3)$$

where $r_f^{C(n)}$ denotes the local risk-free rate for stock n . Note that r_n is independent of numeraire, so that it is the same for all global investors.

The currency component in (15.2) is given by

$$q_n^{\$} = r_f^{C(n)} + r_{FX}^{C(n),\$} - r_f^{\$}. \quad (15.4)$$

This term represents the excess return in US dollars because of holding cash denominated in the local currency of the stock.

We adopt a multi-factor framework to explain the local excess returns. This approach yields valuable insight into the underlying sources of portfolio return by separating systematic effects from the purely stock-specific component that can be diversified away. More specifically, we posit that the local excess returns are driven by a relatively small number K_E , of global equity factors, plus an idiosyncratic component unique to the particular stock,

$$r_n = \sum_{k=1}^{K_E} X_{nk} f_k + u_n. \quad (15.5)$$

Here, X_{nk} ($k \leq K_E$) is the exposure of stock n to equity factor k , f_k is the factor return, and u_n is the specific return of the stock. The specific returns u_n are assumed to be uncorrelated with the factor returns. The factor exposures are known at the start of each period, and the factor returns are estimated via cross-sectional regression.

Suppose that there are K_C currencies in the model. Ordering the currencies after the equity factors, we can express the excess currency returns as

$$q_n^{\$} = \sum_{k=K_E+1}^{K_E+K_C} X_{nk} f_k^{\$}, \quad (15.6)$$

where X_{nk} ($k > K_E$) is the exposure of stock n to currency k , and $f_k^{\$}$ is the excess return of the currency with respect to US dollars (which we calculate from risk-free

rates and exchange rates, as in (15.4)). We take the currency exposures X_{nk} to be equal to 1 if k corresponds to the local currency of stock n , and 0 otherwise.

Equations (15.2), (15.5), and (15.6) can be combined to obtain

$$R_n^{\$} = \sum_{k=1}^K X_{nk} f_k^{\$} + u_n, \quad (15.7)$$

where $K = K_E + K_C$ is the total number of combined equity and currency factors. The elements of X_{nk} define the $N \times K$ factor exposure matrix, where N is the total number of stocks. Note that the factor exposure matrix is independent of numeraire.

Our treatment thus far has considered only a single asset. Our primary objective, however, is to forecast portfolio risk. The portfolio excess return (in US dollars) is given by

$$R_P^{\$} = \sum_{n=1}^N h_n^P R_n^{\$}, \quad (15.8)$$

where h_n^P is the portfolio weight of asset n . Note that, in general, the assets include both stocks and cash. The portfolio exposure to factor k is given by the weighted average of the asset exposures,

$$X_k^P = \sum_{n=1}^N h_n^P X_{nk}. \quad (15.9)$$

In order to estimate portfolio risk, we also require the factor covariance matrix and the specific risk forecasts. The elements of the factor covariance matrix are

$$F_{kl}^{\$} = \text{cov} \left(f_k^{\$}, f_l^{\$} \right), \quad (15.10)$$

where the dollar superscript indicates that at least some of the matrix elements depend on the base currency of the investor. Note, however, that the $K_E \times K_E$ block of the factor covariance matrix corresponding to equity factors is independent of numeraire.

The covariance of specific returns is

$$\Delta_{mn} = \text{cov} (u_m, u_n). \quad (15.11)$$

For most stocks, we assume that the specific returns are uncorrelated, so that the off-diagonal elements of Δ_{mn} are zero. However, (15.11) represents the generalized case in which the specific returns of some securities are linked.²

² We relax the assumption of uncorrelated specific returns for different share classes of the same stock.

The portfolio risk in US dollars can now be obtained as the square root of the variance,

$$\sigma(R_P^{\$}) = \left[\sum_{kl} X_k^P F_{kl}^{\$} X_l^P + \sum_{mn} h_m^P \Delta_{mn} h_n^P \right]^{1/2}. \quad (15.12)$$

Although (15.12) is written for portfolio risk, it is equally valid for tracking error by replacing portfolio weights and exposures with their active counterparts.

Many global investors, of course, are interested in risk forecasts from different numeraire perspectives. The only term on the right-hand-side of (15.12) that depends on base currency is the factor covariance element, $F_{kl}^{\$}$. However, These elements can be transformed to some other numeraire, γ . Substituting these transformed factor covariances F_{kl}^{γ} into (15.12) yields risk forecasts with respect to the new base currency γ .

15.3 Factor Exposures

15.3.1 Estimation Universe

The coverage universe is the set of all securities for which the model provides risk forecasts. The estimation universe, by contrast, is the subset of stocks that is used to estimate the model. Judicious selection of the estimation universe is a critical component in building a sound risk model. The estimation universe must be sufficiently broad to accurately represent the investment opportunity set of global investors, without being so broad as to include illiquid stocks that may introduce spurious return relationships into the model. Furthermore, the estimation universe should be reasonably stable to ensure that factor exposures are well behaved across time. *Representation*, *liquidity*, and *stability*, therefore, represent the three primary goals that must be attained when selecting a risk model estimation universe.

A well-constructed equity index must address and overcome these very issues and therefore serves as an excellent foundation for the estimation universe. The GEM2 estimation universe utilizes the MSCI *All Country World Investable Market Index* (ACWI IMI), part of the MSCI Global Investable Market Indices family which represents the latest in MSCI index-construction methodology. MSCI ACWI IMI aims to reflect the full breadth of global investment opportunities by targeting 99% of the float-adjusted market capitalization in 48 developed and emerging markets. The index-construction methodology applies innovative rules designed to achieve index stability, while reflecting the evolving equity markets in a timely fashion. Moreover, liquidity screening rules are applied to ensure that only investable stocks with reliable pricing are included for index membership.

If a country is deemed excessively restrictive to foreign investment, it is excluded from MSCI ACWI IMI. Nevertheless, many of these frontier markets represent important opportunities for certain global investors. For instance, since 2002, Qualified Foreign Institutional Investors (QFIIs) are permitted to invest directly in the

Chinese domestic A-share market, which does not constitute part of MSCI ACWI IMI. Similarly, the six Gulf Cooperation Council (GCC) markets (Saudi Arabia, Oman, Qatar, UAE, Kuwait, and Bahrain) are important to many global investors, though they are also excluded from MSCI ACWI IMI. We therefore supplement the estimation universe with stocks from China Domestic and the six GCC countries. We down-weight these markets in the regression in order to not distort the factor relationships within the investable universe.

In June 2002, the MSCI Global Investable Market Indices instituted the transition to float-adjusted market-cap weighting and the application of additional liquidity screening rules. In this month, the number of constituents in MSCI ACWI IMI dropped sharply from nearly 12,000 to below 7,000. Although this corresponds to a drop of only 5% on a cap-weighted basis, this event must be treated carefully in the estimation universe to avoid potentially spurious jumps in factor exposures. We remove the discontinuity in the estimation universe by backward-excluding all stocks that were dropped from MSCI ACWI IMI in June 2002. We allow exceptions to this rule in cases where additional stocks are needed to populate countries with few assets.

The official MSCI ACWI IMI history begins in June 1994. To obtain a deeper model history, however, we backward-extend the estimation universe for 2 years by including the constituents of the standard MSCI *All Country World Index*, which contains large-cap and mid-cap stocks. We supplement this universe with all US and Japanese small-cap stocks with MSCI ACWI IMI membership as of June 1994. While this does lead to a jump in the number of estimation universe constituents in June 1994, it does not lead to jumps in risk forecasts since it occurs before the first available factor covariance matrix.

15.3.2 GEM2 Factor Structure

The equity factor set in GEM2 includes a World factor (w), countries (c), industries (i), and styles (s). Every stock is assigned an exposure of 1 to the World factor. Hence, the local excess returns in (15.5) can be rewritten as

$$r_n = f_w + \sum_c X_{nc} f_c + \sum_i X_{ni} f_i + \sum_s X_{ns} f_s + u_n. \quad (15.13)$$

Mathematically, the World factor represents the intercept term in the cross-sectional regression. Economically, it describes the aggregate up-and-down movement of the global equity market. Typically, the World factor is the dominant source of risk for a diversified long-only portfolio.

For most institutional investors, however, the primary concern is the risk of *active* long/short portfolios. If both the portfolio and benchmark are fully invested – as is typically the case – then the active exposure to the World factor is zero. Similarly, if the long and short positions of a long/short portfolio are of equal absolute value, the net exposure to the World factor is zero. Thus, we must look beyond the World factor to other sources of risk.

Country factors play a critical role in global equity risk modeling. One reason is that they are powerful indicator variables for explaining the cross section of global equity returns. A second, related, reason is that the country allocation decision is central to many global investment strategies, and portfolio managers often must carefully monitor their exposures to these factors. We therefore include explicit country factors for all markets covered.

In Table 15.1, we present a list of the 55 countries covered by GEM2, together with their corresponding currencies. The country exposures X_{nc} in GEM2 are set equal to 1 if stock n is in country c , and set equal to 0 otherwise. We assign country exposures based on country membership within the MSCI ACWI IMI, MSCI China A Index and MSCI GCC Countries Index. Note that depository receipts and cross-listed assets are assigned factor exposures for the underlying or primary asset, as defined by the MSCI Equity Indices.

We also show in Table 15.1 the average and ending weights (based on total market capitalization) over the period, from January 1997 to January 2008. It is interesting to note the relative decline of the US and Japanese markets, and the rise of markets such as Brazil, Russia, India, and China. We also report the number of securities in the estimation universe for each country and the market capitalization (float-adjusted and total) in billions of US Dollars as of January 2008.

Industries are also important variables in explaining the sources of global equity return co-movement. One of the major strengths of GEM2 is to employ the Global Industry Classification Standard (GICS[®]) for the industry factor structure. The GICS scheme is hierarchical, with 10 top-level sectors, which are then divided into 24 industry groups, 68 industries, and 154 sub-industries. GICS applies a consistent global methodology to classify stocks based on careful evaluation of the firm's business model and economic operating environment. The GICS structure is reviewed annually by MSCI Barra and Standard & Poor's to ensure that it remains timely and accurate.

Identifying which industry factors to include in the model involves a combination of judgment and empirical analysis. At one extreme, we could use the 10 GICS sectors as industry factors. Such broad groupings, however, would certainly fail to capture much of the cross-sectional variation in stock returns. At the other extreme, we could use all 154 sub-industries as the factor structure. Besides the obvious difficulties associated with the unwieldy numbers of factors (e.g., risk reporting, thin industries), such an approach would present a more serious problem for risk forecasting: although adding more factors always increases the in-sample R^2 of the cross-sectional regressions, many of the factor returns would not be statistically significant. Allowing noise-dominated "factors" into the model defeats the very purpose of a factor risk model.

In GEM2, selection of the industry factor structure begins at the second level of the GICS hierarchy, with each of the 24 industry groups automatically qualifying as a factor. This provides a reasonable level of granularity, without introducing an excessive number of factors. We then analyze each industry group by carefully examining the industries and sub-industries contained therein to determine if a more granular factor structure is warranted. The basic criteria, we use to guide

Table 15.1 GEM2 country factors and currencies

Country code	Country name	Currency name	Average weight	Jan-08 weight	Jan-08 # stocks	Jan-08 float-cap	Jan-08 total-cap
ARG	Argentina	Argentine Peso	0.10	0.09	13	19.43	41.67
AUS	Australia	Australian Dollar	1.62	2.48	236	1045.32	1197.26
AUT	Austria	Euro	0.17	0.37	35	94.68	176.79
BHR	Bahrain (*)	Bahraini Dinar	0.01	0.02	11	3.13	10.40
BEL	Belgium	Euro	0.64	0.69	47	195.07	333.81
BRA	Brazil	Brazilian Real	0.65	1.87	140	531.80	904.92
CAN	Canada	Canadian Dollar	2.59	3.44	297	1372.74	1658.38
CHL	Chile	Chilean Peso	0.18	0.26	34	48.63	126.19
CHN	China Domestic (*)	Chinese Yuan	1.90	7.00	1511	1209.86	3379.20
CHX	China International	Hong Kong Dollar	0.67	2.96	215	637.21	1429.87
COL	Colombia	Colombian Peso	0.03	0.07	11	13.38	32.18
CZE	Czech Republic	Czech Koruna	0.06	0.15	8	28.85	74.13
DNK	Denmark	Danish Krone	0.39	0.46	57	153.77	220.54
EGY	Egypt	Egyptian Pound	0.05	0.17	31	33.44	81.72
FIN	Finland	Euro	0.59	0.69	52	281.29	334.69
FRA	France	Euro	4.18	5.24	184	1611.22	2528.75
DEU	Germany	Euro	3.31	3.71	184	1431.50	1790.54
GRC	Greece	Euro	0.28	0.48	62	136.76	229.30
HKG	Hong Kong	Hong Kong Dollar	1.07	1.50	165	395.80	723.88
HUN	Hungary	Hungarian Forint	0.06	0.09	8	29.21	41.89
IND	India	Indian Rupee	0.54	2.41	251	382.40	1164.89
IDN	Indonesia	Indonesian Rupiah	0.13	0.34	54	74.93	164.14
IRE	Ireland	Euro	0.26	0.27	30	112.05	128.76
ISR	Israel	Israeli Shekel	0.20	0.32	69	89.30	152.88
ITA	Italy	Euro	2.05	2.09	161	633.60	1009.46
JPN	Japan	Japanese Yen	11.52	8.96	1160	3198.33	4322.65
JOR	Jordan	Jordanian Dinar	0.03	0.05	17	4.43	23.73
KOR	Korea	Korean Won	1.02	2.06	332	604.46	993.49

(continued)

Table 15.1 (continued)

Country code	Country name	Currency name	Average weight	Jan-08 weight	Jan-08 # stocks	Jan-08 float-cap	Jan-08 total-cap
KWT	Kuwait (*)	Kuwaiti Dinar	0.14	0.33	85	72.41	157.53
MYS	Malaysia	Malaysian Ringgit	0.43	0.53	119	108.21	255.87
MEX	Mexico	Mexican Peso	0.43	0.59	40	172.93	285.52
MAR	Morocco	Moroccan Dirham	0.03	0.09	10	10.40	45.13
NLD	Netherlands	Euro	1.47	1.34	70	445.70	645.96
NZL	New Zealand	New Zealand Dollar	0.09	0.07	22	24.77	33.57
NOR	Norway	Norwegian Krone	0.34	0.71	68	186.40	344.12
OMN	Oman (*)	Omani Rial	0.01	0.03	14	5.92	15.84
PAK	Pakistan	Pakistan Rupee	0.03	0.06	24	8.35	29.97
PER	Peru	Peruvian Sol	0.04	0.11	16	25.06	54.67
PHL	Philippines	Philippine Peso	0.06	0.14	27	21.78	67.20
POL	Poland	Polish Zloty	0.11	0.33	66	73.29	157.75
PRT	Portugal	Euro	0.20	0.24	21	57.70	117.57
QAT	Qatar (*)	Qatari Rial	0.05	0.17	23	17.25	83.92
RUS	Russia	Russian Ruble	0.49	2.83	71	306.42	1364.70
SAU	Saudi Arabia (*)	Saudi Rial	0.32	0.75	40	169.49	363.13
SGP	Singapore	Singapore Dollar	0.48	0.72	104	186.03	348.62
ZAF	South Africa	South African Rand	0.63	0.86	116	293.06	415.28
ESP	Spain	Euro	1.41	2.11	88	664.97	1018.36
SWE	Sweden	Swedish Krone	1.05	1.05	105	389.92	506.61
CHE	Switzerland	Swiss Franc	2.54	2.41	118	1021.22	1163.85
TWN	Taiwan	Taiwan Dollar	1.25	1.34	411	436.94	646.55
THA	Thailand	Thailand Bhat	0.19	0.36	66	59.72	174.41
TUR	Turkey	New Turkish Lira	0.18	0.45	68	72.55	214.91
GBR	UK	U.K. Pound	8.65	7.65	463	3456.81	3694.66
ARE	UAE (*)	Emirati Dirham	0.06	0.43	40	44.77	206.47
USA	US	US Dollar	47.53	34.80	2468	15742.12	16794.26
Total			102.50	108.74	10138	38446.78	52482.54

Weights are normalized excluding China Domestic and GCC (marked by asterisk), and computed within the GEM2 estimation universe using total market capitalization. Average weights are from January 1997 to January 2008. Market capitalizations are reported in billions of US Dollars

industry factor selection, are: (a) the groupings of industries into factors must be economically intuitive, (b) the industry factors should have a strong degree of statistical significance, (c) incorporating an additional industry factor should significantly increase the explanatory power of the model, and (d) thin industries (those with few assets) should be avoided.

The result of this process is the set of 34 GEM2 industry factors, presented in Table 15.2. Industries that qualify as factors tend to exhibit volatile returns and have significant weight. We find that this relatively parsimonious set of factors captures

Table 15.2 GEM2 Industry Factors

GICS sector	GEM2 code	GEM2 industry factor name	Average weight	Jan-08 weight
Energy	1	Energy equipment & services	0.75	1.29
	2	Oil, gas & consumable fuels	4.88	9.32
	3	Oil & gas exploration & production	1.00	1.72
Materials	4	Chemicals	2.36	2.84
	5	Construction, containers, paper	1.38	1.24
	6	Aluminum, diversified metals	1.05	2.41
	7	Gold, precious metals	0.37	0.58
	8	Steel	0.79	1.83
Industrials	9	Capital goods	7.33	8.60
	10	Commercial & professional services	1.43	0.77
	11	Transportation Non-Airline	1.82	2.32
	12	Airlines	0.37	0.45
Consumer discretionary	13	Automobiles & components	2.52	2.29
	14	Consumer durables & apparel	2.33	1.93
	15	Consumer services	1.35	1.39
	16	Media	3.24	2.11
	17	Retailing	3.42	2.08
Consumer staples	18	Food & staples retailing	1.82	1.76
	19	Food, beverage & tobacco	4.56	4.37
	20	Household & personal products	1.43	1.20
Health care	21	Health care equipment & services	2.13	1.93
	22	Biotechnology	0.78	0.68
	23	Pharmaceuticals, life sciences	6.17	3.82
Financials	24	Banks	10.52	10.83
	25	Diversified financials	5.63	5.06
	26	Insurance	4.61	4.14
	27	Real estate	2.08	3.07
Information Technology	28	Internet software & services	0.62	0.74
	29	IT services, software	3.24	2.56
	30	Communications equipment	2.46	1.41
	31	Computers, electronics	3.69	2.81
Telecom	32	Semiconductors	2.47	1.52
	33	Telecommunication services	7.11	5.84
Utilities	34	Utilities	4.31	5.08

Weights are computed within the GEM2 estimation universe using total market capitalization. Average weights are computed from January 1997 to January 2008

most of the in-sample R^2 explained by the 154 sub-industries, but with a much higher degree of statistical significance. Also reported in Table 15.2 are the average and end-of-period industry weights from January 1997 to January 2008. The weights were computed using the entire GEM2 estimation universe (i.e., including China Domestic and GCC countries). Only five industries have end-of-period weights less than 100 bps, and these tend to be highly volatile, thus making them useful risk factors.

Investment style represents another important source of systematic risk. Style factors, also known as *risk indices*, are designed to capture these sources of risk. They are constructed from financially intuitive stock attributes called *descriptors*, which serve as effective predictors of equity return covariance. Since the descriptors within a particular style factor are meant to capture the same underlying driver of returns, these descriptors tend to be significantly collinear. For instance, price-to-book ratio, dividend yield, and earnings yield are all attributes used to identify value stocks, and they tend to exhibit significant cross-sectional correlation. Although these descriptors have significant explanatory power on their own, naively including them as separate factors in the model may lead to serious multi-collinearity problems. Combining these descriptors into a single style factor overcomes this difficulty and also leads to a more parsimonious factor structure.

Unlike country and industry factors, which are assigned exposures of either 0 or 1, style factor exposures are continuously distributed. To facilitate comparison across style factors, they are standardized to have a mean of 0 and a standard deviation of 1. Each descriptor is also standardized similarly. That is, if d_{nl}^{Raw} is the raw value of stock n for descriptor l , then the standardized descriptor value is given by

$$d_{nl} = \frac{d_{nl}^{Raw} - \mu_l}{\sigma_l}, \quad (15.14)$$

where μ_l is the cap-weighted mean of the descriptor (within the estimation universe), and σ_l is the equal-weighted standard deviation. We adopt the convention of standardizing using the cap-weighted mean so that a well-diversified cap-weighted global portfolio, such as MSCI ACWI IMI, has approximately zero exposure to all style factors. For the standard deviation, however, we use the equal-weighted mean to prevent large-cap stocks from having an undue influence on the overall scale of the exposures.

Formally, descriptors are combined into risk indices as follows

$$X_{nk} = \sum_{l \in k} w_l d_{nl}, \quad (15.15)$$

where w_l is the descriptor weight, and the sum takes place over all descriptors within a particular risk index. Descriptor weights are determined using an optimization algorithm to maximize the explanatory power of the model. Style factor exposures are rescaled to have a standard deviation of 1.

Some of the style factors are standardized on a *global-relative* basis, others on a *country-relative* basis. In the former case, the mean and standard deviation in (15.14) are computed using the entire global cross section. In the latter case, the

factors have mean 0 and standard deviation 1 within each country. When deciding which standardization convention to adopt, we consider both the intuitive meaning of the factor and its explanatory power.

GEM2 uses eight style factors. Below, we provide a qualitative description of each of the style factors:

- The *Volatility* factor is typically the most significant style factor. In essence, it captures market risk that cannot be explained by the World factor. The most important descriptor within the Volatility index is historical beta relative to the World portfolio (as proxied by the estimation universe). To better understand this factor, consider a fully invested long-only portfolio that is strongly tilted toward high-beta stocks. Intuitively, this portfolio has greater market risk than a portfolio with beta equal to one. This additional market risk is captured through positive exposure to the Volatility factor. Note that the time-series correlation between the World factor and the Volatility factor is typically very high, so that these two sources of risk add coherently in this example. If, by contrast, the portfolio is invested in low-beta stocks, then the risk from the Volatility and the World factors is partially canceled, as intuitively expected. We standardize the Volatility factor on a global-relative basis. As a result, the mean exposure to Volatility within a country can deviate significantly from zero. This standardization convention is a natural one for a global model, as most investors regard stocks in highly volatile markets as having more exposure to the factor than those in low-volatility markets. This view is reflected in the data, as we find that the explanatory power of the factor is greater using the global-relative standardization.
- The *Momentum* factor often ranks second in significance after Volatility. Momentum differentiates stocks based on recent relative performance. Descriptors within Momentum include historical alpha from a 104-week regression and relative strength (over trailing 6 and 12 months) with a 1-month lag. Similarly to Volatility, Momentum is standardized on a global-relative basis. This is also an intuitive convention for a global model. From the perspective of a global investor, a stock that strongly outperforms the World portfolio is likely to be considered a positive momentum stock, even if it slightly underperforms its country peers. The empirical results support this view, as the Momentum factor standardized globally has greater explanatory power than one standardized on a country-relative basis.
- The *Size* factor represents another well-known source of return covariance. It captures the effect of large-cap stocks moving differently from small-cap stocks. We measure Size by a single descriptor: log of market capitalization. The explanatory power of the model is quite similar whether Size is standardized globally or on a country-by-country basis. We adopt the country-relative standardization, however, as it is more intuitive and consistent with investors' perception of the markets. For instance, major global equity indices, such as the MSCI Global Investable Market Indices, segment each country according to size, with the largest stocks inside each country always being classified as large-cap stocks. Moreover, standardizing the Size factor on a global-relative basis would serve as an unintended proxy for developed markets versus emerging markets, and increases collinearity with the country factors.

- The *Value* factor describes a major investment style which seeks to identify stocks that are priced low relative to fundamentals. We standardize Value on a country-relative basis. This again is consistent with the way major indices segment each market, with each country divided roughly equally into value and growth sub-indices. This convention also circumvents the difficulty of comparing fundamental data across countries with different accounting standards. GEM2 utilizes official MSCI data items for Value factor descriptors, as described in the *MSCI Barra Fundamental Data Methodology Handbook* (2008).
- *Growth* differentiates stocks based on their prospects for sales or earnings growth. It is standardized on a country-relative basis, consistent with the construction of the MSCI Value and Growth Indices. Therefore, each country has approximately half the weight in stocks with positive Growth exposure, and half with negative exposure. The GEM2 Growth descriptors also utilize official MSCI data items, as described in the *MSCI Barra Fundamental Data Methodology handbook*.
- The *Non-Linear Size (NLS)* factor captures non-linearities in the payoff to size exposure across the market-cap spectrum. NLS is based on a single raw descriptor: the cube of the log of market capitalization. Since this raw descriptor is highly collinear with the Size factor, we orthogonalize it to the Size factor. This procedure does not affect the fit of the model, but does mitigate the confounding effects of collinearity and thus preserves an intuitive meaning for the Size factor. The NLS factor is represented by a portfolio that goes long mid-cap stocks and shorts large-cap and small-cap stocks.
- The *Liquidity* factor describes return patterns to stocks based on relative trading activity. Stocks with high turnover have positive exposure to Liquidity, whereas low-turnover stocks have negative exposure. Liquidity is standardized on a country-relative basis.
- *Leverage* captures the return difference between high-leverage and low-leverage stocks. The descriptors within Leverage include market leverage, book leverage, and debt-to-assets ratio. This factor is standardized on a country-relative basis.

In Appendix A, we provide additional details on the individual descriptors comprising each style factor.

15.4 Factor Returns

15.4.1 Data Quality and Outlier Treatment

The most difficult and time-consuming part of constructing a global equity risk model lies in the preparation of the input data. If the data inputs are garbage, the risk forecasts will likewise be garbage, no matter how sophisticated the model. Assuring a high degree of data quality therefore is a vital part of building a reliable risk model.

In order to obtain the highest quality inputs, GEM2 leverages the same data infrastructure used for the construction of MSCI Global Investable Market Indices.

Data items such as raw descriptors, GICS codes, country classifications, and clean daily stock returns are obtained from in-house sources that have already undergone extensive quality control.

No matter how stringent the data quality assurance process, we can never exclude the possibility of extreme outliers entering the data set. These extreme outliers may represent legitimate values or outright data errors. Either way, such observations must be carefully handled to prevent a few data points from having an undue impact on model estimation.

In GEM2, we employ a multi-step algorithm to identify and treat outliers. The algorithm assigns each observation into one of three groups. The first group represents values so extreme that they are treated as potential data errors and removed from the estimation process. The second group represents values that are regarded as legitimate, but nonetheless so large that their impact on the model must be limited. We *trim* these observations to three standard deviations from the mean. The third group of observations, which forms the bulk of the distribution, consists of values less than three standard deviations from the mean. These observations are left unadjusted.

The thorough data quality assurance process, coupled with the robust outlier algorithm, ensures that the input data are clean and reliable. There is still the issue, however, of dealing with *missing* data. Data could be missing either due to the lack of availability or due to the removal by the outlier algorithm. The data items most likely to be missed include descriptors for the Value, Growth, Leverage, and Liquidity style factors. More rarely, stocks may be missing exposure to Volatility and Momentum factors; this occurs for securities which lack return history, such as recent spin-offs or IPOs. Other data items, such as Size, industry, and country exposures, are never missing, since these items are required for coverage universe membership.

If the underlying data required to compute factor exposures are missing, then the factor exposures are generated using a data-replacement algorithm. A simple approach would be to assign zero exposure to all missing values. A better approach, however, is to exploit known relationships between factor exposures. For instance, if we know that a stock is in the Semiconductor industry and has a positive exposure to Volatility, then it is likely that the stock also has a positive exposure to Growth.

We apply the data-replacement algorithm to the Value, Growth, Leverage, and Liquidity factors. The algorithm works by regressing these four factors (using only non-missing exposures) against Size, Volatility and industries. The slope coefficients of the regression are then used to estimate the factor exposures for the stocks with missing data. Countries are not used as explanatory variables in the regression since the four style factors (Value, Growth, Leverage and Liquidity) are standardized to be mean zero with respect to countries. The algorithm is not applied to the other four style factors (Size, NonLinear Size, Volatility and Momentum) since these factor exposures are rarely, if ever, missing. Note also that the algorithm is applied at the factor level as opposed to the descriptor level. In other words, if a stock has data for some descriptors within a risk index, but not others, then the non-missing data will be used to compute the factor exposures. Only if *all* descriptors are absent does the replacement algorithm become active.

15.4.2 GEM2 Regression Methodology

The equity factor returns f_k in GEM2 are estimated by regressing the local excess returns r_n against the factor exposures X_{nk} ,

$$r_n = \sum_{k=1}^{K_E} X_{nk} f_k + u_n. \quad (15.16)$$

GEM2 uses weighted least squares, assuming that the variance of specific returns is inversely proportional to the square root of total market capitalization.

As described in Section 15.3, the GEM2 equity factors include the World factor, countries, industries, and styles. Every stock in GEM2 has unit exposure to the World factor, and indicator variable exposures of 0 or 1 to countries and industries. As a result, the sum of all country factors equals the World factor, and similarly for industries, i.e.,

$$\sum_c X_{nc} = 1, \text{ and } \sum_i X_{ni} = 1, \quad (15.17)$$

for all stocks n . In other words, the sum of all country columns in the factor exposure matrix gives a column with 1 in every entry, which corresponds to the World factor. The same holds for industry factors. The GEM2 factor structure therefore exhibits exact twofold collinearity. Constraints must be applied to obtain a unique solution.

In GEM2, we adopt constraints as in [Heston and Rouwenhorst \(1994\)](#) that require the cap-weighted country and industry factor returns to sum to zero,

$$\sum_c w_c f_c = 0, \quad \text{and} \quad \sum_i w_i f_i = 0, \quad (15.18)$$

where w_c is the weight of the estimation universe in country c , and w_i is the corresponding weight in industry i . These constraints remove the exact collinearities from the factor exposure matrix, without reducing the explanatory power of the model.

We can now give a more precise interpretation to the factors. Consider the cap-weighted estimation universe, with holdings h_n^E . The return of this portfolio R_E can be attributed using the GEM2 factors,

$$R_E = f_w + \sum_c w_c f_c + \sum_i w_i f_i + \sum_s X_s^E f_s + \sum_n h_n^E u_n. \quad (15.19)$$

The constraints imply that the first two sums in (15.19) are equal to zero. The third sum is also zero since the style factors are standardized to be cap-weighted mean zero; i.e., $X_s^E = 0$, for all styles s . The final sum in (15.19) corresponds to the specific return of a broadly diversified portfolio and is *approximately* zero (note it would be *exactly* zero if we used regression weights instead of capitalization weights). Thus, to an excellent approximation, (15.19) reduces to

$$R_E \approx f_w. \quad (15.20)$$

In other words, the return of the World factor is essentially the cap-weighted return of the estimation universe.

To better understand the meaning of the pure factor portfolios, we report in Table 15.3 the long and short weights (as of January 2008) of the World portfolio and several pure factor portfolios in various market segments. The World portfolio is represented by the cap-weighted GEM2 estimation universe. The pure World factor is 100% net long and the net weights closely match those of the World portfolio in each segment. The other pure factors have net weight of zero and therefore represent long/short portfolios.

As a first approximation, the pure country factors can be regarded as going long 100% the particular country, and going short 100% the World portfolio. For instance, going long 100% Japan and short 100% the World portfolio results in a portfolio with roughly 91% weight in Japan, and -91% in all other countries. The pure country factors, however, have zero exposure to industry factors. This is accomplished by taking appropriate long/short combinations in other countries. For instance, the Japanese market is over-represented in the segment corresponding to the Automobile factor. To partially hedge this exposure, the pure Japan factor takes a net short position of -1.09% in the US Automobile segment. A similar short position would be found in the German Automobile segment.

The pure Automobile factor can be thought, as a first approximation, to be formed by going 100% long the Automobile industry and 100% short the World portfolio. A more refined view of the factor takes into account that the net weight in each country is zero. The pure Automobile factor naturally takes a large long position in Japanese automobiles, but hedges the Japan exposure by taking short positions in other Japanese segments.

The pure Volatility factor is perhaps the easiest to understand, as it takes off-setting long and short positions within all segments corresponding to GEM2 factors (e.g., Japan, US, and Automobiles). Note that the weights are not equal to zero for segments that do not correspond to GEM2 factors, such as Japanese automobiles.

15.4.3 Characteristics of GEM2 Factors

In this section, we present and discuss some of the quantitative characteristics of the GEM2 factors. In particular, we investigate the degree of collinearity among the factor exposures and report on the statistical significance, performance, and volatility of the factor returns.

A feature of the multi-factor framework is that it disentangles the effects of many variables acting simultaneously. Multi-collinearity among factors, however, can confound this clean separation of effects.

One measure of collinearity is the pair-wise cross-sectional correlation between factor exposures. In Table 15.4, we report regression-weighted correlations among style factors and industries, averaged over the period from January

Table 15.3 Segment weights, as of January 2008, for the World portfolio and several pure factor portfolios

Segment	(ESTU) World portfolio	Pure World factor	Pure Japan factor	Pure US factor	Pure Auto factor	Pure Volatility factor
World (Net)	100.00	100.00	0.00	0.00	0.00	0.00
Long	100.00	106.61	97.09	66.41	112.61	54.91
Short	0.00	-6.61	-97.09	-66.41	-112.61	-54.91
Japan (Net)	8.96	8.96	91.04	-8.96	0.00	0.00
Long	8.96	10.00	91.04	0.18	33.22	5.79
Short	0.00	-1.05	0.00	-9.14	-33.22	-5.79
US (Net)	34.80	34.80	-34.80	65.20	0.00	0.00
Long	34.80	34.99	1.55	65.21	19.59	16.37
Short	0.00	-0.19	-36.34	0.00	-19.59	-16.37
Auto (Net)	2.33	2.33	0.00	0.00	97.67	0.00
Long	2.33	2.45	4.31	0.96	97.67	1.16
Short	0.00	-0.12	-4.31	-0.96	0.00	-1.16
Japan Auto (Net)	1.05	0.87	4.31	-0.06	33.21	0.18
Long	1.05	0.89	4.31	0.05	33.21	0.38
Short	0.00	-0.03	0.00	-0.11	0.00	-0.20
US Auto (Net)	0.21	0.27	-1.09	0.89	15.51	0.22
Long	0.21	0.27	0.00	0.89	15.51	0.32
Short	0.00	0.00	-1.09	0.00	0.00	-0.10

The World portfolio is represented by the cap-weighted GEM2 estimation universe

1997 to June 2008. In general, the correlations are intuitive in sign. For instance, Volatility is positively correlated with Growth, Liquidity, and Semiconductors, and negatively with Value and Utilities. Value, as expected, is positively correlated with Banks and Utilities, and negatively with Growth and Biotechnology. Although none of the correlations are particularly large, the correlations between pairs of style factors are typically larger, on average, than those between styles and industries.

The statistical significance of factor returns plays a key role in the construction of a risk model. Let f_k be the factor return for a particular period, and $se(k)$ be the standard error of this estimate. The t -statistic is given by

$$t_k = \frac{f_k}{se(k)}. \quad (15.21)$$

Generally, a t -statistic with absolute value greater than 2 is considered significant at the 95% confidence level. Ideally, a risk factor should have high t -statistics that are persistent across time. Useful measures of this are the average squared t -statistic over time, and the percentage of observations with significant t -statistics.

In Table 15.5, we report the summary statistics for the GEM2 style factors. In this case, we divide the sample period into two sub-periods. In the first sample period (from January 1997 to June 2002), the style factors were generally more volatile and had a greater degree of statistical significance than in the second sub-period (July 2002 to June 2008). This is not too surprising, since the first sub-period contains the rise and fall of the internet bubble. Note that the kurtosis of the Value factor in the second sample period is particularly high. This is caused primarily by 2 six-sigma events in August 2007.

We can discern from Table 15.5 a clear risk hierarchy in the style factors. Alone at the top is the Volatility factor, which is far more volatile and significant than any of the other factors. In the second tier are Momentum and Size, which are clearly set apart from the third group, containing the other five styles. In this third group, the Value factor tends to be the most significant.

Performance, however, tells a different story. Volatility performed very poorly during the first sub-period and was essentially flat during the second. Value and Momentum, on the other hand, were strongly positive over both sample periods.

There is substantial correlation between some of the style factors with the estimation universe. For instance, from Table 15.5, we see that Volatility is strongly correlated (about 80%) with the estimation universe. This makes sense since during up-markets, high-beta stocks tend to outperform low-beta stocks, so that the return to Volatility – which includes beta – is also positive. A similar argument holds for down-markets. Liquidity and Size also tend to be positively correlated with the estimation universe over both sample periods.

Although not reported here, the country and industry factor returns are highly significant. The mean average squared t -statistic is 31.7 for countries and 17.2 for industries over the sample period from January 1997 to June 2008. Also, more than half of the observations are statistically significant at the 95% confidence level.

Table 15.4 Regression-weighted cross-sectional correlation of style and industry factor exposures

Factor	Volatility	Momentum	Size	Value	Growth	NL Size	Liquidity	Leverage
Volatility	1.00			−0.25	0.27	−0.06	0.42	−0.07
Momentum	−0.05	1.00	−0.09	−0.17	0.08	0.15	0.09	−0.08
Size	−0.09	0.12	1.00	−0.11	−0.05	0.11	0.10	−0.03
Value	−0.25	−0.17	−0.11	1.00	−0.24	−0.03	−0.15	0.22
Growth	0.27	0.08	−0.05	−0.24	1.00	−0.02	0.18	−0.17
Non-linear Size	−0.06	0.15	0.11	−0.03	−0.02	1.00	0.11	0.01
Liquidity	0.42	0.09	0.10	−0.15	0.18	0.11	1.00	−0.03
Leverage	−0.07	−0.08	−0.03	0.22	−0.17	0.01	−0.03	1.00
Energy equipment & services	0.06	0.02	−0.03	−0.02	0.06	0.02	0.07	−0.03
Oil, gas & consumable fuels	−0.02	0.02	0.11	0.06	−0.03	−0.04	0.00	−0.01
Oil & gas exploration & production	0.01	0.02	−0.01	0.01	0.00	0.00	0.02	0.01
Chemicals	−0.03	0.00	−0.01	0.04	−0.04	0.03	−0.01	0.00
Construction, containers, paper	−0.04	−0.01	−0.02	0.07	−0.06	0.04	−0.04	0.06
Aluminum, diversified metals	0.03	0.02	0.02	0.03	0.00	0.01	0.03	0.00
Gold, precious metals	0.00	−0.01	0.00	−0.06	0.00	0.00	−0.01	−0.04
Steel	0.04	0.02	−0.01	0.09	−0.02	0.01	0.03	0.03
Capital goods	0.01	0.01	−0.07	0.04	−0.02	0.01	0.00	−0.01
Commercial & professional services	0.01	0.00	−0.09	−0.05	0.06	−0.01	−0.02	−0.03
Transportation non-airline	−0.06	0.01	0.00	0.02	−0.02	0.02	−0.04	0.08
Airlines	0.02	−0.02	0.01	0.02	−0.01	0.02	0.02	0.08
Automobiles & components	0.01	0.00	0.01	0.10	−0.02	0.00	−0.01	0.00
Consumer durables & apparel	0.01	−0.01	−0.07	0.05	0.00	0.00	0.03	−0.06
Consumer services	−0.02	0.00	−0.06	−0.03	0.04	0.02	0.01	0.02
Media	−0.01	−0.03	0.01	−0.12	0.02	0.02	−0.03	0.00
Retailing	0.03	0.01	−0.05	−0.01	0.05	0.01	0.04	−0.06
Food & staples retailing	−0.07	−0.01	0.02	−0.02	0.00	0.01	−0.03	−0.01

(continued)

Table 15.4 (continued)

Factor	Volatility	Momentum	Size	Value	Growth	NL Size	Liquidity	Leverage
Food, beverage & tobacco	-0.14	0.00	0.02	-0.01	-0.05	0.02	-0.07	-0.01
Household & personal products	-0.05	0.00	0.03	-0.04	-0.02	-0.01	-0.03	-0.03
Health care equipment & services	-0.02	0.02	-0.06	-0.08	0.09	0.01	0.04	-0.05
Biotechnology	0.09	0.00	-0.05	-0.17	0.07	-0.03	0.07	-0.05
Pharmaceuticals, life sciences	-0.04	0.00	0.06	-0.10	0.02	-0.04	0.00	-0.10
Banks	-0.08	0.00	0.12	0.12	-0.09	-0.05	-0.10	0.10
Diversified financials	0.05	0.00	0.02	0.04	0.01	0.01	-0.03	0.04
Insurance	-0.06	0.00	0.04	0.11	-0.05	0.03	-0.06	-0.10
Real estate	-0.13	0.02	-0.09	0.07	-0.08	0.03	-0.07	0.17
Internet software & services	0.15	-0.02	-0.04	-0.12	0.12	-0.03	0.07	-0.06
IT services, software	0.15	-0.01	-0.06	-0.14	0.12	-0.02	0.09	-0.12
Communications equipment	0.16	-0.02	-0.01	-0.09	0.06	-0.03	0.08	-0.07
Computers, electronics	0.11	-0.02	-0.02	-0.05	0.04	-0.03	0.09	-0.09
Semiconductors	0.21	-0.02	0.00	-0.10	0.09	-0.02	0.16	-0.08
Telecommunication services	0.04	-0.02	0.15	-0.07	0.01	-0.07	0.01	0.05
Utilities	-0.17	0.01	0.06	0.11	-0.13	0.04	-0.07	0.20

Results are averages over the period January 1997 to June 2008. Correlations above 0.10 in absolute value are shaded in gray.

Table 15.5 Style factor summary statistics based on weekly factor returns

Factor name	Average squared <i>t</i> -statistic	Percent observations with $ t > 2$	Factor kurtosis (weekly)	Annualized factor return	Annualized factor volatility	Factor Sharpe ratio	Correlation with ESTU
<i>A. January 1997 to June 2002 (66 months)</i>							
Volatility	119.46	82.6	4.92	-5.98	8.21	-0.73	0.79
Momentum	48.12	78.5	6.34	7.62	4.46	1.71	0.11
Size	43.65	79.9	4.73	2.37	2.97	0.80	0.29
Value	11.76	57.6	4.55	7.65	1.96	3.89	-0.16
Growth	9.75	50.7	4.49	-0.09	1.61	-0.05	0.41
Non-linear Size	8.86	49.7	4.31	1.61	1.89	0.85	0.15
Liquidity	10.18	53.8	4.00	4.99	1.63	3.06	0.51
Leverage	4.73	33.0	5.55	-1.18	1.13	-1.04	-0.03
Average	32.06	60.72	4.86	2.12	2.98	1.06	0.26
<i>B. July 2002 to June 2008 (72 months)</i>							
Volatility	89.50	84.8	4.07	0.36	4.98	0.07	0.84
Momentum	35.37	76.4	5.71	4.62	2.65	1.75	-0.11
Size	26.45	65.0	5.33	0.71	1.67	0.42	0.41
Value	11.89	53.1	10.74	3.86	1.37	2.83	0.08
Growth	4.45	30.4	4.86	1.00	0.80	1.24	0.04
Non-linear Size	6.85	44.0	5.34	1.25	1.11	1.13	0.17
Liquidity	8.46	46.3	4.34	0.35	1.25	0.28	0.46
Leverage	4.30	33.7	8.59	-0.09	0.84	-0.11	0.02
Average	23.41	54.21	6.12	1.51	1.83	0.95	0.24

The sample period, January 1997 to June 2008, is divided into two sub-periods, A and B

Finally, it is worth noting that the time-series correlation between the World factor and the cap-weighted estimation universe return is 99.4%, thus confirming our interpretation of the World factor.

15.4.4 Comparing the Explanatory Power of the Models

An important summary statistic used to measure the explanatory power of a model is the R^2 of the cross-sectional regressions,

$$R^2 = 1 - \frac{\sum_n w_n u_n^2}{\sum_n w_n r_n^2}, \quad (15.22)$$

where w_n is the regression weight of stock n , u_n is the specific return, and r_n is the local excess return. The R^2 can be interpreted as the ratio of the average *explained* squared return to the average *total* squared return.

In Fig. 15.1, we plot the trailing 12-month R^2 of GEM2 factor groups over time. The sample period is from January 1994 to June 2008. The first case we examine is the World factor in isolation. This is the single most important factor and, on average, explains about 10% in R^2 . Its explanatory power, however, fluctuates over time. For instance, in 2000 the World factor explained only a few percentage points, whereas this figure rose to nearly 20% by 2003.

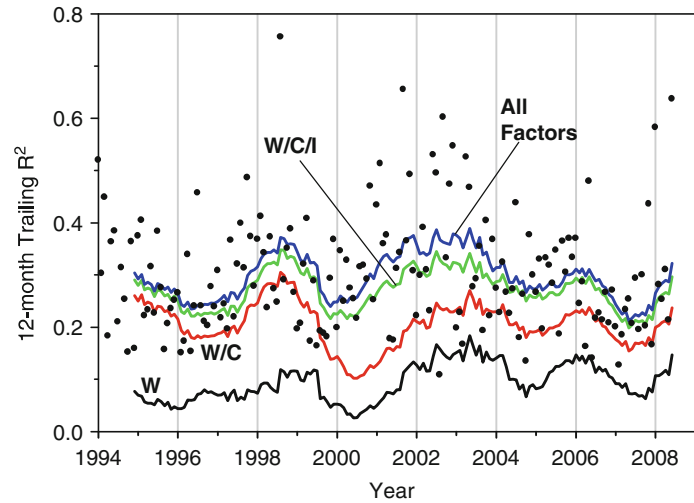


Fig. 15.1 Explanatory power of GEM2 factor groups, expressed as trailing 12-month R^2 . Here, W indicates the World factor in isolation, W/C denotes World factor plus countries, and $W/C/I$ represents World, country, and industry factors. The *black dots* indicate the actual monthly R^2 values. Over the sample period January 1994 to June 2008, the average R^2 values are approximately 10% (W), 21% (W/C), 27% ($W/C/I$) and 30% (all factors)

From Fig. 15.1, it is evident that most of the explanatory power of the model comes from the combined effect of country and industry factors. Interestingly, the *relative* explanatory power of these factor groups changes significantly over time. In the early period, industries explained relatively little compared with countries. Then, some time in 1999, the explanatory power of industries suddenly exploded. From 2000 to 2003, industries actually explained a greater proportion of trailing 12-month R^2 than countries. Since 2003, the situation has partially reversed, with country factors reasserting their dominance though industry factors remained very important. The average R^2 obtained using the World factor with country factors is about 21%, whereas adding industry factors on top of the countries increases the R^2 to about 27%. In other words, country factors add about 11% in R^2 above the World factors, and industries add another 6%.³ This example illustrates some of the insights obtained by the GEM2 factor structure, which places countries and industries on an equal footing.

The style factors contribute about 300 bps in R^2 beyond that explained by countries and industries alone. This amount, however, also varies considerably over time. In the early period, style factors added only about 100 bps, whereas this widened to about 500 bps during the 2000–2004 period. More recently, style factors have added about 200 bps.

The black dots in Fig. 15.1 represent the monthly R^2 values for the complete set of GEM2 factors. This quantity varies dramatically from month to month, and tends to be largest during big market moves. In extreme months, commonalities in returns swamp asset-specific returns. For example, the largest single R^2 observed during the sample period was 0.75. This occurred in August 1998, when the market dropped by more than 15%.

15.5 GEM2 Factor Covariance Matrix

The GEM2 factor covariance matrix is built from the time series of weekly factor returns, f_{kt} . Weekly observations reduce sampling error and provide more responsive risk forecasts than monthly observations. Daily factor returns are not used for computing global covariances due to asynchronous trading effects, but they are computed for performance attribution purposes.

We use exponentially weighted moving averages (EWMA) to estimate the factor covariance matrix for both equity and currency blocks. This approach gives more weight to recent observations and is a simple, yet robust, method for dealing with data non-stationary. An alternative approach would be to use generalized auto-regressive conditional heteroskedasticity, or GARCH. We find that EWMA estimators are typically more robust to changing market dynamics and produce more accurate risk forecasts than their GARCH counterparts.

³ We obtain similar results if industry factors are introduced before country factors.

In GEM2, we must also account for the possibility of serial correlation in factor returns, which can affect risk forecasts over a longer horizon. Suppose, for instance, that high-frequency returns are negatively correlated. In this case, long-horizon risk forecasts estimated on high-frequency data will be lower than that implied using simple square-root-of-time scaling, since returns one period tend to be partially offset by opposing returns the following period.

The prediction horizon in Barra risk models is typically 1 month. Models that are estimated on daily or weekly returns therefore must adjust for serial correlation. Note that models estimated on monthly observations need not adjust for serial correlation, since the observation frequency coincides with the prediction horizon.

Full treatment of serial correlation must account for not only the correlation of one factor with itself across time, but also for the correlation of two factors with each other across different time periods. In GEM2, we model serial correlation using the methodology of [Newey and West \(1987\)](#) with two lags. This assumes that the return of any factor may be correlated with the return of any other factor up to 2 weeks prior.

It is useful to think of the factor covariance matrix as being composed of a correlation matrix, which is then scaled by the factor volatilities. Volatilities and correlations can then be estimated separately using EWMA with different half-life parameters.

The volatility half-life determines the overall responsiveness of the model. Selecting the proper volatility half-life involves analyzing the trade-off between accuracy and responsiveness on the one hand, and stability on the other. If the volatility half-life is too long, the model gives undue weight to distant observations that have little to do with current market conditions. This leads to stable risk forecasts but at the cost of reduced accuracy. By contrast, a short volatility half-life makes the model more responsive, generally improving the bias statistics, but at the cost of jumpier risk forecasts. Of course, the volatility half-life cannot be lowered without bound; if the half-life is too short, then sampling error can become so large that the risk forecasts are not only less stable, but also less accurate.

We measure the accuracy of the risk forecasts using bias statistics, which essentially represent the ratio of realized risk to forecast risk. A bias statistic of 1 is considered ideal, but sampling error ensures that realized bias statistics will deviate from 1, even for perfect risk forecasts. In Appendix B, we provide a review of bias statistics.

When carrying out bias statistic testing, one must consider the number of months to include in the observation window. Sampling error suggests that one should use the largest interval possible, since that would minimize sampling error. This would be a compelling argument for stationary data. However, in reality financial markets are non-stationary. It is possible to over-predict risk of several years and then under-predict for others, while obtaining a bias statistic close to 1. Getting the right *average* risk forecast over 10 or 20 years is small consolation to a portfolio manager who can be wiped out due to poor risk forecasts in any single year.

Rolling 12-month intervals provide a more relevant framework for evaluating the accuracy of risk forecasts. Although sampling error is larger, this approach penalizes

the model for poor risk forecasts during any individual year. For a given portfolio, we compute the rolling absolute deviation (RAD) of the bias statistic as follows. First, we calculate the bias statistic over the first year of the observation window, using 12 monthly observations. We then take the absolute value of the deviation of the bias statistic from 1. We then roll the 12-month window forward 1 month at a time, computing absolute deviations each month, until we reach the end of the observation window. For instance, a 138-month observation window will accommodate 127 separate 12-month windows. The RAD for the portfolio is the average of the absolute deviations over these 127 sub-periods. The RAD measure captures in a single number the accuracy of risk forecasts over rolling 12-month intervals by penalizing all observations throughout the sample history that deviate from the ideal bias statistic of 1.

As shown in Appendix B, for perfect risk forecasts and normally distributed returns (kurtosis 3), the critical value of the RAD over a 138-month observation window is 0.22. In other words, if the observed RAD is larger than 0.22, then with 95% confidence, we can reject the null hypothesis of an accurate risk forecast. Returns, of course, are not normally distributed. For a kurtosis of 5, which corresponds to slightly fat tails, the critical value of RAD for a 138-month observation window is 0.29. For even fatter tails, say a kurtosis of 10, the critical value increases to 0.34.

To study the question of the stability of risk forecasts, we compute the monthly absolute change in volatility for each factor k ,

$$v_k(t) = \frac{|\sigma_k(t+1) - \sigma_k(t)|}{\sigma_k(t)}. \quad (15.23)$$

If the volatility forecast changes over 1 month from 6.0 to 5.7%, for instance, then the monthly absolute change is 5%.

The *mean variability* provides a measure of the stability of the risk forecasts and is given by the average of the monthly absolute changes,

$$\bar{v} = \frac{1}{KT} \sum_{k,t} v_k(t), \quad (15.24)$$

where K is the number of factors and T is the number of time periods.

In Fig. 15.2, we show the mean RAD for all GEM2 equity factors, plotted as a function of volatility half-life. The bias statistics were computed over rolling 12-month windows using a 138-month observation window from January 1997 to June 2008. We also plot the mean monthly variability $\bar{v}$ of the equity factors. The volatility half-life of the GEM2 model is indicated by the vertical line.

The solid lines in Fig. 15.2 are the results which incorporate a two-lag autocorrelation adjustment, as in GEM2. For comparison, we also show the corresponding results by dashed lines without serial correlation adjustments. Including autocorrelation adjustments makes the risk forecasts more accurate but also slightly increases the monthly variability in the forecasts.

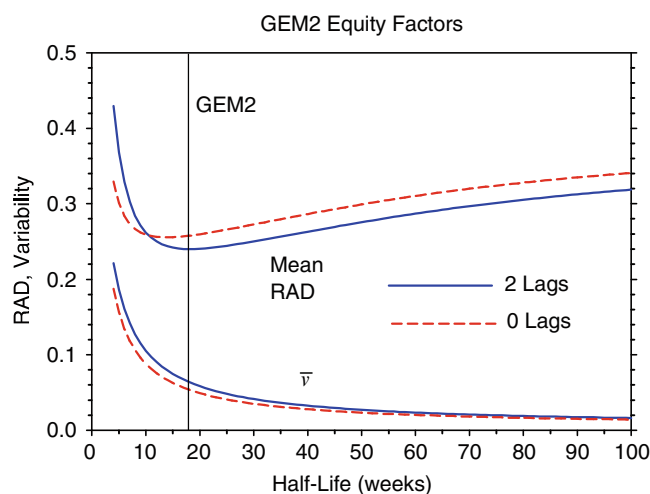


Fig. 15.2 Mean 12-month rolling absolute deviation (RAD) and mean Variability ($\bar{v}$) of GEM2 equity factors, plotted as a function of volatility half-life. The sample period was 138 months from January 1997 to June 2008. The solid lines indicate results with the two-lag autocorrelation that is used GEM2, and the dashed lines are for no autocorrelation adjustment. The vertical line indicates the actual GEM2 half-life parameter

From Fig. 15.2, we see that the volatility half-life of GEM2 (18 weeks) minimizes the mean RAD and thus produces the most accurate risk forecasts. Some users, however, place a premium on the *stability* of the risk forecast. In this case, a longer half-life may be desirable even though the risk forecasts are slightly less accurate. For instance a 52-week volatility half-life produces a mean RAD of about 0.27 (vs. 0.24 for GEM2), but the monthly variability is reduced to 2.5% (vs. 6.7% for the 18-week half-life). This example illustrates the tradeoff between accuracy and stability.

Figure 15.2 also clearly demonstrates the effect of sampling error. Namely, as the half-life is reduced below about 15 weeks, the RAD increases dramatically while the variability of the risk forecast also increases.

We also investigated how the accuracy and stability of *currency* risk forecasts varied with volatility half-life. Although not shown here, the results are qualitatively and quantitatively almost identical to the results reported in Fig. 15.2. Namely, the 18-week volatility half-life minimizes the mean RAD (at 0.26), and produces a mean variability of 6.6%. A half-life of 52 weeks produces an RAD of 0.30 but has more stable risk forecasts with a mean variability of 2.6%. The effect of autocorrelation for currencies is also comparable to that shown in Fig. 15.2.

Special care must be exercised when selecting the correlation half-life. To ensure a well-conditioned correlation matrix, the number of observations T must be significantly greater than the number of factors K . In the extreme case that $T < K$, the correlation matrix will have zero eigenvalues. This can present problems for portfolio optimization, as it would be possible to find spurious active portfolios with seemingly zero factor risk.

For GEM2, we use a correlation half-life of 104 weeks. A useful rule of thumb is that the number of effective observations is roughly three times the half-life. Therefore, a correlation half-life of 104 weeks corresponds to roughly 300 effective observations. The volatility and correlation half-lives used in GEM2 are the same as those used in other Barra single-country models, such as USE3S.

Another complication in estimating the factor covariance matrix arises for the case of missing factor returns. One possible reason for missing data is holidays. For instance, stocks do not trade in China during Golden Week, so it is not possible to directly estimate a return to the China Domestic factor. More commonly, however, missing factor returns arise from using time series of differing lengths. For instance, the GCC stocks appear in the estimation universe as of July 2000, so the factor returns prior to this date are missing.

We use the EM algorithm of [Dempster, Laird, and Rubin \(1977\)](#) to estimate the factor covariance matrix in the presence of missing factor returns. This method employs an iterative procedure to estimate the covariance matrix, conditional on all available data. The EM algorithm assures that the factor covariance matrix is positively semi-definite.

15.6 Specific Risk

15.6.1 GEM2 Specific Risk Model

GEM2 uses a structural model to estimate specific risk. This methodology is similar to that used in other Barra risk models, but employs weekly rather than monthly observations. These higher-frequency observations allow us to make the forecasts more responsive while also reducing estimation error.

Mathematically, the specific risk forecast of a stock is given as the product of three components

$$\sigma_n = \hat{S} \left(1 + \hat{V}_n \right) K_M, \quad (15.25)$$

where $\hat{S}$ is the forecast mean absolute specific return of stocks in the estimation universe, $\hat{V}_n$ is the forecast relative absolute specific return, and K_M is a multiplicative scalar which depends on the market-cap segment M of the stock. The overall responsiveness of the model is dominated by $\hat{S}$, whereas $\hat{V}_n$ determines the cross-sectional variation in specific risk. The role of K_M is to convert from weekly absolute specific returns to monthly standard deviation. The motivation for modeling specific risk in terms of absolute specific returns is that the estimation process is less prone to outliers.

The *realized* mean absolute specific return for a given period t is computed as the cap-weighted average,

$$S_t = \sum_n w_{nt} |u_{nt}|, \quad (15.26)$$

where u_{nt} is the realized specific return for stock n . We use the cap-weighted average because large-cap assets are less prone to extreme outliers. The forecast mean absolute specific return is given by the exponentially weighted average of the realized values,

$$\hat{S} = \sum_t \gamma_t S_t. \quad (15.27)$$

For GEM2, we use a half-life parameter of 9 weeks for γ_t . To determine this half-life parameter, we first constructed a broad group of active portfolios. We then selected the half-life parameter γ_t so that the specific risk forecast variability of these portfolios roughly matched the variability in forecasts of the factors, while also taking into account model performance. This results in highly accurate specific risk forecasts and greater stability in the proportion of risk coming from factors and stock-specific components.

In Fig. 15.3, we plot the realized mean weekly absolute specific return over the period from June 1992 to June 2008, together with the forecast values from GEM2. Overall, the model does an excellent job forecasting the absolute specific return levels. In the mid-1990s, the absolute specific returns were quite low (about 2%), but then began to rise, hitting a peak of nearly 6% in early 2000. After the collapse of the internet bubble, absolute specific return levels fell sharply over the next 5 years. Since 2007, however, specific risk levels have again begun to rise.

We use a factor model to forecast the relative absolute specific return, whose *realized* values are given by

$$\varepsilon_{nt} = \frac{|u_{nt}| - S_t}{S_t}. \quad (15.28)$$

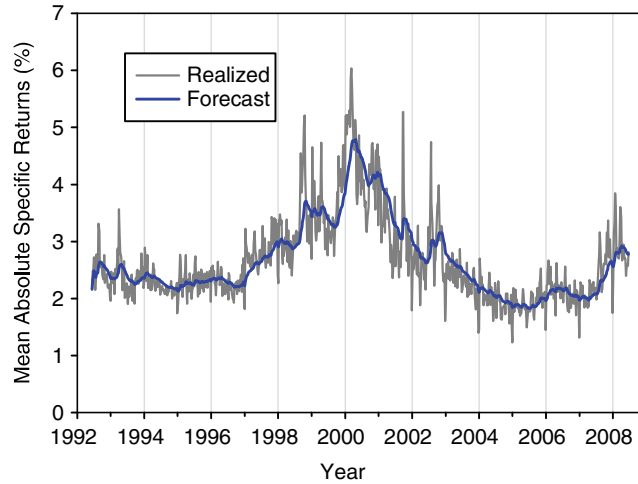


Fig. 15.3 Plot of weekly realized and forecast mean absolute specific returns for the GEM2 estimation universe

GEM2 factors already provide a sound basis for explaining ε_{nt} , since the level of relative absolute specific risk depends on the country, industry, and style exposures of the individual stocks. For instance, large-cap utility stocks tend to have lower specific returns than, say, small-cap internet stocks.

While the model factors are well suited to forecast specific risk, the most powerful explanatory variable is simply the trailing realized specific volatility. We therefore augment the GEM2 factor exposure matrix with this additional factor for the purpose of forecasting the relative absolute specific returns.

To estimate the relationship between factor exposures and relative absolute specific returns, we perform a 104-week pooled cross-sectional regression,

$$\varepsilon_{nt} = \sum_k \tilde{X}_{nk}^t g_k + \lambda_{nt}, \quad (15.29)$$

where $\tilde{X}_{nk}^t$ represents the augmented factor exposure matrix element. We use capitalization weights in the regression to reduce the impact of high-kurtosis small-cap stocks and apply exponential weighting with a half-life of 26 weeks. The forecast relative absolute specific return is given by

$$\hat{V}_n = \sum_k \tilde{X}_{nk} g_k, \quad (15.30)$$

where $\tilde{X}_{nk}$ is the most recent factor exposure matrix, and g_k is the slope coefficient estimated over the trailing 104 weeks.

The product $\hat{S}(1 + \hat{V}_n)$ is the weekly forecast absolute specific return for the stock. We need to convert this to a monthly standard deviation. If the specific returns were normally distributed (temporally), then the time-series standard deviation of the ratio

$$\frac{u_{nt}}{S_t(1 + \hat{V}_{nt})} \quad (15.31)$$

would be $\sqrt{\pi/2}$, where $\hat{V}_{nt}$ is the forecast relative absolute specific return at time t . However, since the specific returns exhibit kurtosis, the standard deviation is usually slightly greater than this theoretical value. For this reason, the multiplicative scalar K_M is sometimes called the *kurtosis correction*.

There is one final adjustment that must be applied to the kurtosis correction. The prediction horizon of the risk model is 1 month. We observe, on average, a small but persistent negative serial correlation in weekly specific returns. As a result, the monthly specific volatility will be slightly less than that suggested by simple square root of time scaling. If we rank stocks by degree of serial correlation, however, we find no relationship between the rank of stocks over non-overlapping periods. We therefore define the standardized return with two-lag autocorrelation,

$$b_{nt} = \frac{u_{n,t} + u_{n,t-1} + u_{n,t-2}}{\sqrt{3}S_t(1 + \hat{V}_{nt})}. \quad (15.32)$$

The kurtosis correction is given by the realized standard deviation of b_{nt} ,

$$K_M = \sigma(b_{nt}), \quad (15.33)$$

where the standard deviation is computed over all observations within a particular market-cap segment M over the trailing 104 weeks.

15.6.2 Specific Risk Bias Statistics

In this section, we evaluate the accuracy of the specific risk forecasts. We consider the entire 138-month sample period and compute rolling 12-month bias statistics for each stock. There are 127 rolling 12-month windows that fit into the sample period. The number of stocks in the estimation universe on average is approximately 8,000. Thus, we compute roughly one million 12-month bias statistics.

The cap-weighted distribution of these bias statistics is shown in Fig. 15.4. The mean of the distribution is 1.02, indicating that on average the specific risk forecasts are essentially unbiased. The cap-weighted mean RAD is approximately 0.23, which is slightly more accurate than the mean RAD of 0.24 that was reported for factor volatility forecasts (see Fig. 15.2). Approximately, 86% of the observations fall within the confidence interval. If returns were normally distributed, then we would expect about 95% of the observations to fall within the confidence interval. In fact, kurtosis values of about 5 are typical of specific returns. In Fig. 15.6 of Appendix B,

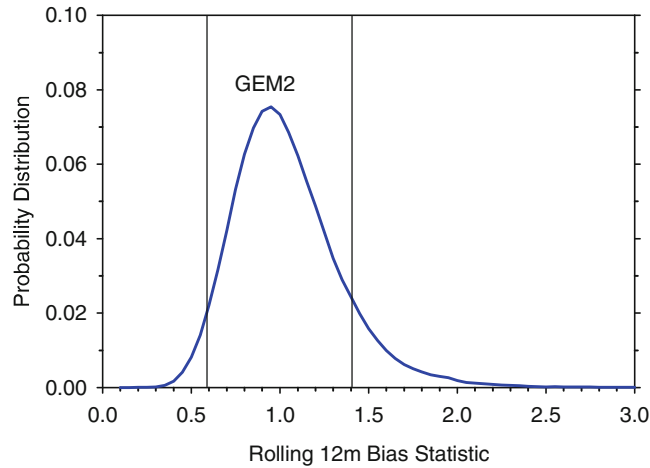


Fig. 15.4 Frequency distribution of 12-month rolling bias statistics for specific risk forecasts. Sample period is January 1997 to June 2008. Results are cap-weighted within the GEM2 estimation universe. Vertical lines indicate confidence interval [0.59, 1.41]

we show that for a kurtosis of 5 over 138 periods, about 86% of the observations should fall within the confidence interval for perfect risk forecasts, consistent with the GEM2 results.

We also computed the distribution of equal-weighted rolling 12-month bias statistics. The results are largely similar to the cap-weighted case. The equal-weighted mean bias statistic is 1.01, and about 82% of the observations fall within the confidence interval. The mean RAD is 0.26 in this case. These results indicate that the GEM2 specific risk forecasts are highly accurate across the entire market-cap spectrum.

15.7 Model Performance

In this section, we evaluate the accuracy of risk forecasts for a combination of country, industry, and style portfolios. The country and industry portfolios were constructed by carving out the constituents of the GEM2 estimation universe from the appropriate segment, and then cap-weighting the stocks. The style portfolios were constructed by ranking all stocks according to their exposures to the particular style factor, and then cap-weighting the stocks that ranked in the top and bottom 20%. For each of the long-only country, industry, and style portfolios thus formed, we also construct long/short active portfolios by selecting MSCI ACWI IMI as the benchmark. The sample period in all cases was from January 1997 to June 2008.

In Fig. 15.5, we plot the distribution of the 12-month rolling bias statistics for the complete set of country, industry, and style *active* portfolios. The sample period

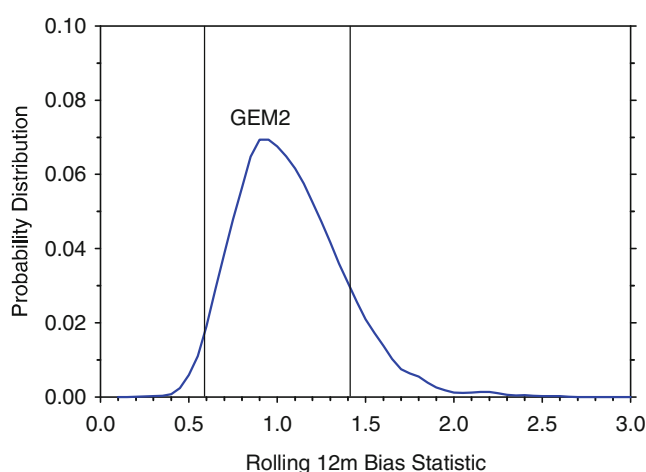


Fig. 15.5 Frequency distribution of 12-month rolling bias statistics for country, industry and style active portfolios (long/short) described in text. Sample period is January 1997 to June 2008. Vertical lines indicate the confidence interval [0.59, 1.41]

was from January 1997 to June 2008. The vertical lines denote the approximate confidence interval assuming normally distributed returns. The mean of the distribution is 1.04, with a mean RAD of 0.24. About 86% of the observations fall within the confidence interval. The kurtosis values of the active portfolios are typically in the range of 4–5. At these kurtosis levels, Fig. 15.6 of Appendix B indicates that for perfect risk forecasts about 86–90% of the observations should fall within the confidence interval, roughly consistent with the GEM2 results.

We also investigated the bias statistic distribution for the same set of active portfolios over two sub-periods. The first sub-period runs from January 1997 to June 2002 (66 months), and corresponds to a period of generally rising volatility. The second sub-period runs from July 2002 to June 2008 (72 months), and corresponds to a period of falling volatility. In the first sub-period, the mean rolling 12-month bias statistic is 1.10. For the second sub-period, it is 0.98. The fact that these results do not deviate far from 1 indicates that the GEM2 model does a good job of adapting to changing market dynamics.

15.8 Conclusion

Forecasting global equity risk is an integral part of any disciplined global investment strategy. In this paper, we presented a detailed description of the Barra global equity model, GEM2. This model incorporates the latest advances in equity risk modeling, including (a) an improved factor structure to better capture the cross section of asset returns, (b) higher frequency observations for reduced sampling error and more responsive forecasts, (c) the introduction of a World factor to cleanly separate the effects of countries and industries, and (d) a structural specific risk model for highly accurate forecasts.

Appendix A: Descriptors by Style Factor

Volatility (Global Relative)

Historical beta (β_n) (HBETA)

Computed as the slope coefficient in a time-series regression of local excess stock returns r_{nt} against the cap-weighted local excess return of the estimation universe R_t ,

$$r_{nt} = \alpha_n + \beta_n R_t + e_{nt}. \quad (15.34)$$

The regression coefficients are estimated on the trailing 104 weeks of returns.

Daily standard deviation (DASTD)

This descriptor differentiates stocks based on recent volatility, and is computed as the standard deviation of daily local returns over the past 65 trading days.

Cumulative range (CMRA)

This descriptor differentiates stocks that have experienced wide swings over the last 12 months from those that have traded in a narrow range. Let $Z(T)$ be the cumulative local excess logarithmic return over the past T months,

$$Z(T) = \sum_{t=1}^T \ln(1 + r_{nt}) - \sum_{t=1}^T \ln(1 + r_{ft}). \quad (15.35)$$

where r_{nt} is the local return of stock n for month t , and r_{ft} is the risk-free return of the local currency. The cumulative range is given by

$$CMRA = \ln(1 + Z_{\max}) - \ln(1 + Z_{\min}), \quad (15.36)$$

where $Z_{\max} = \max \{Z(T)\}$, $Z_{\min} = \min \{Z(T)\}$, and $T = 1, \dots, 12$.

Historical sigma (σ_n) (HSIGMA)

Computed as the volatility of residual returns in (15.34),

$$\sigma_n = \text{std}(e_{nt}). \quad (15.37)$$

The volatility is estimated over the trailing 104 weeks of returns.

Momentum (Global Relative)*Historical alpha (α_n) (HALPHA)*

Given by the intercept term α_n in (15.34). Regression coefficients are estimated on trailing 104 weeks of returns.

Relative strength (6 months) (RSTR6)

Computed as the cumulative local excess logarithmic return over the previous 6 months,

$$RSTR6 = \sum_{t=1}^6 \ln(1 + r_{nt}) - \sum_{t=1}^6 \ln(1 + r_{ft}), \quad (15.38)$$

with a 1-month lag. Here, r_{nt} is the local return of stock n for month t , and r_{ft} is the risk-free return of the local currency.

Relative strength (12 months) (RSTR12)

Computed as the cumulative local excess logarithmic return over the previous 12 months,

$$RSTR12 = \sum_{t=1}^{12} \ln(1 + r_{nt}) - \sum_{t=1}^{12} \ln(1 + r_{ft}), \quad (15.39)$$

with a 1-month lag.

Size (Country Relative)*Log of market cap (LNCAP)*

Given by the logarithm of the total market capitalization of the firm.

Value (Country Relative)*Predicted earnings-to-price ratio (EPFWD)*

Given by the 12-month forward-looking earnings per share (*EPS12F*) divided by the current price. Forward-looking earnings per share are defined as a weighted average between the mean analyst-predicted earnings per share for the current and next fiscal years. For details, refer to Section 1.3 of *MSCI Barra Fundamental Data Methodology*.

Book-to-price ratio (BTOP)

Computed using the latest book value per share and the price. For details, refer to Section 1.2.5 of *MSCI Barra Fundamental Data Methodology*.

Cash earnings-to-price ratio (CETOP)

Computed using the trailing 12-month cash earnings per share (*CEPS*) divided by the current price. For details, refer to Section 1.2.5 of *MSCI Barra Fundamental Data Methodology*.

Trailing earnings-to-price ratio (ETOP)

Given by the trailing 12-month earnings per share (*EPS*) divided by the current price. For details on the calculation of the *EPS*, refer to Section 2.1 of *MSCI Barra Fundamental Data Methodology*.

Dividend-to-price ratio (YIELD)

Given by the annualized dividend-per-share (*DPS*) divided by the current price. For details, refer to Section 1.2.4 of *MSCI Barra Fundamental Data Methodology*.

Growth (Country Relative)*Long-term predicted earnings growth (EGRLF)*

Long-term (3–5 years) earnings growth rate forecasted by analysts. For details, refer to Section 2.2.5 of *MSCI Barra Fundamental Data Methodology*.

Sales growth (trailing 5 years) (SGRO)

Annual reported sales per share are regressed against time over the past 5 fiscal years. The slope coefficient is then divided by the average annual sales per share to obtain the sales growth. For details, refer to Section 2.2.1 of *MSCI Barra Fundamental Data Methodology*.

Earnings growth (trailing 5 years) (EGRO)

Annual reported earnings per share are regressed against time over the past 5 fiscal years. The slope coefficient is then divided by the average annual earnings per share to obtain the earnings growth. For details, refer to Section 2.2.1 of *MSCI Barra Fundamental Data Methodology*.

Non-linear Size (Country Relative)

Cube of log of market cap (NLSIZE)

First, the standardized size exposure (i.e., log of market cap) is cubed. The resulting factor is then orthogonalized to the size factor on a regression-weighted basis. The orthogonalized factor is then winsorized and standardized.

Liquidity (Country Relative)

Average share turnover, trailing 12 months (STOA)

Computed as the logarithm

$$STOA = \log \left(\frac{1}{12} \sum_t \frac{V_t}{S_t} \right), \quad (15.40)$$

where V_t is the trading volume of the asset for month t , and S_t is the number of shares outstanding. The sum runs over the past 12 months.

Average share turnover, trailing 3 months (STOQ)

Computed as the logarithm

$$STOQ = \log \left(\frac{1}{3} \sum_t \frac{V_t}{S_t} \right), \quad (15.41)$$

where V_t is the trading volume of the asset for month t , and S_t is the number of shares outstanding. The sum runs over the past 3 months.

Share turnover, 1 month (STOM)

Computed as the logarithm

$$STOM = \log \left(\frac{V_t}{S_t} \right), \quad (15.42)$$

where V_t is the trading volume of the asset for the month, and S_t is the corresponding number of shares outstanding.

Leverage (Country Relative)

Market leverage (MLEV)

Computed as

$$MLEV = \frac{MCAP + PREF + LD}{MCAP}, \quad (15.43)$$

where $MCAP$ is the market value of common equity at previous month-end, $PREF$ is the most recent book value of preferred equity, and LD is the most recent book value of long-term debt. For details, refer to Section 15.5 of *MSCI Barra Fundamental Data Methodology*.

Book leverage (BLEV)

Computed as

$$BLEV = \frac{BV + PREF + LD}{BV}, \quad (15.44)$$

where BV is the most recent book value of common equity, $PREF$ is the most recent book value of preferred equity, and LD is the most recent book value of long-term debt. For details, refer to Section 15.5 of *MSCI Barra Fundamental Data Methodology*.

Debt-to-assets (DTOA)

Computed as

$$DTOA = \frac{TD}{TA}, \quad (15.45)$$

where TD is the book value of total debt (long-term debt, LD , and current liabilities, CL), and TA is most recent book value of total assets. For details, refer to Section 15.5 of *MSCI Barra Fundamental Data Methodology*.

Appendix B: Review of Bias Statistics

B1. Single-Window Bias Statistics

To assess a model's predictive performance, it is not enough to consider its explanatory power; we must also test how well its risk forecasts perform out of sample. In this section, we describe how to evaluate the accuracy of risk forecasts.

A commonly used measure to assess a risk model's accuracy is the bias statistic. The bias statistic, conceptually, represents the ratio of realized risk to forecast risk.

Let r_{nt} be the return to portfolio n over period t , and let σ_{nt} be the beginning-of-period volatility forecast. Assuming perfect forecasts, the *standardized* return,

$$b_{nt} = \frac{r_{nt}}{\sigma_{nt}}, \quad (15.46)$$

has expected standard deviation 1. The bias statistic for portfolio n is the *realized* standard deviation of standardized returns,

$$B_n = \sqrt{\frac{1}{T-1} \sum_{t=1}^T (b_{nt} - \bar{b}_n)^2}, \quad (15.47)$$

where T is the number of periods in the observation window.

Assuming normally distributed returns and perfect risk forecasts, for sufficiently large T the bias statistic B_n is approximately normally distributed about 1, and roughly 95% of the observations fall within the confidence interval,

$$B_n \in \left[1 - \sqrt{2/T}, 1 + \sqrt{2/T}\right]. \quad (15.48)$$

If B_n falls outside this interval, then we reject the null hypothesis that the risk forecast was accurate.

If returns are not normally distributed, however, then fewer than 95% of the observations will fall within the confidence interval, *even for perfect risk forecasts*. In Fig. 15.6, we show simulated results for the percentage of observations actually

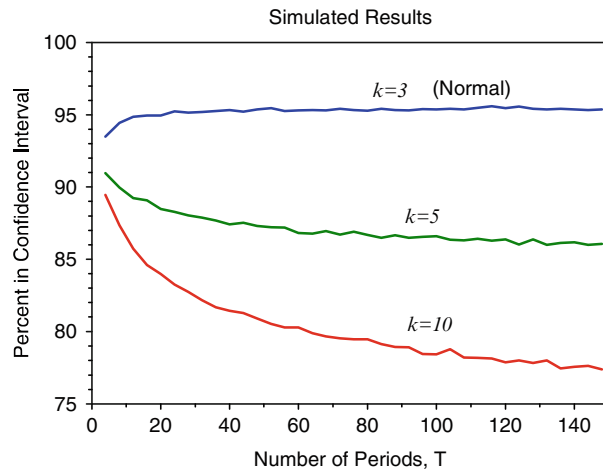


Fig. 15.6 Percent of observations falling within confidence interval $\left[1 - \sqrt{2/T}, 1 + \sqrt{2/T}\right]$, where T is the number of periods in the observation window. Results were simulated using a normal distribution ($k = 3$), and using a t -distribution with kurtosis values $k = 5$ and $k = 10$. The standard deviations were equal to 1 in all cases (i.e., perfect risk forecasts). For the normal distribution, the percentage of observations inside the confidence interval quickly approaches 95%. As kurtosis is increased, however, the proportion within the confidence interval declines considerably

falling within this interval, plotted versus observation-window length T , for several values of kurtosis k .

For the normal case (kurtosis $k = 3$), except for the smallest values of T , the confidence interval indeed captures about 95% of the observations. As the kurtosis increases, however, the percentage falling within the interval drops significantly. For instance, even for a fairly modest kurtosis level of 5, only 86% of bias statistics fall inside the confidence interval for an observation window of 120 periods.

B2. Rolling-Window Bias Statistics

The purpose of bias-statistic testing is to assess the accuracy of risk forecasts, typically over a long sample period. Let T be the length of the observation window, which corresponds to the number of months in the sample period. One possibility is to select the entire sample period as a single window, and to compute the bias statistic as in (15.47). This would be a good approach if financial data were stationary, as sampling error is reduced by increasing the length of the window. In reality, however, financial data are not stationary. It is possible to significantly over-predict risk for some years, and under-predict it for others, while ending up with a bias statistic very close to 1.

A more relevant question is to study how accurate the risk forecasts were during any 12-month period. For this purpose, we define the rolling 12-month bias statistic for portfolio n ,

$$B_n^\tau = \sqrt{\frac{1}{11} \sum_{t=\tau}^{\tau+11} (b_{nt} - \bar{b}_n)^2}, \quad (15.49)$$

where τ denotes the first month of the 12-month window. The 12-month windows are rolled forward 1 month at a time until reaching the end of the observation window. If T is the number of periods in the observation window, then each portfolio will have $T - 11$ (overlapping) 12-month windows. It is often informative to consider the frequency distribution of rolling 12-month bias statistics. These histograms quickly indicate whether there were any 12-month periods for which portfolio risk was significantly over-forecast or under-forecast. Figures 15.4 and 15.5 are examples of such distributions. If there are N portfolios, then the histogram will contain $N(T - 11)$ data points.

It is also useful to consider the mean of the rolling 12-month bias statistics, given by

$$\tilde{B} = \frac{1}{N(T - 11)} \sum_{n, \tau} B_n^\tau. \quad (15.50)$$

This number indicates whether, on average, risk was over-forecast or under-forecast during the observation window.

Again, it is possible to over-forecast the risk of some portfolios within certain time periods, and to under-forecast the risk of other portfolios during different time periods, while producing a mean rolling bias statistic $\tilde{B}$ close to 1. What is needed is a way to penalize every deviation away from the ideal bias statistic of 1. The 12-month *rolling absolute deviation* (RAD) for portfolio n , defined as

$$\text{RAD}_n = \frac{1}{T-11} \sum_{\tau=1}^{T-11} |B_n^\tau - 1|, \quad (15.51)$$

captures this effect. Smaller RAD numbers, of course, are preferable to larger ones. To assess whether an observed RAD is statistically significant or not, we must consider the properties of the RAD distribution.

In Fig. 15.7, we plot the probability distributions for 12-month RAD over observation windows of 12, 36, and 150 months. Results were obtained via numerical simulation using normally distributed returns. The plot labeled by $T = 12$ corresponds to a single period. The most likely RAD occurs near zero in this case, and the mean of the distribution is approximately $\sqrt{1/12\pi}$, or 0.163. A more precise computation, which accounts for small sample sizes, gives a mean of 0.17.

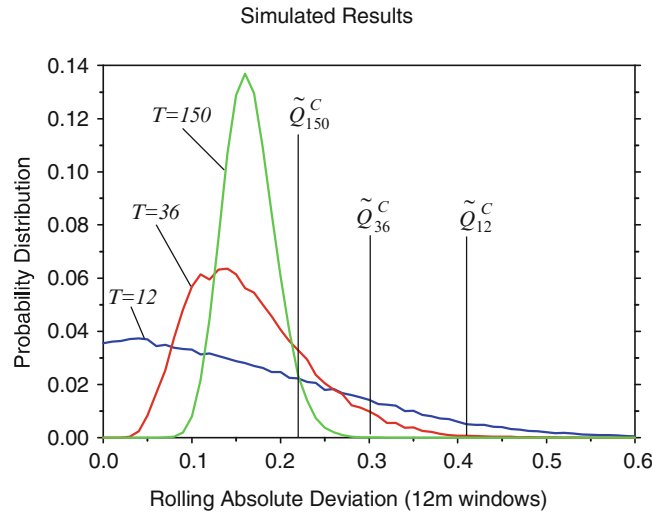


Fig. 15.7 Probability distribution of 12-month rolling absolute deviation (RAD). Results were simulated using normally distributed returns, for *observation* windows of 12, 36, and 150 months. Since a 12-month rolling window within a 12-month observation window corresponds to a single fixed window, the curve labeled $T = 12$ is peaked near zero. A 36-month observation window, however, corresponds to roughly three independent observations, and the peak of the distribution shifts to the right. As the number of periods T becomes large, the central limit theorem assures that the distribution converges to a normal distribution centered at approximately 0.17. Critical values $\tilde{Q}_T^C$ are indicated by the vertical lines. They indicate the 95th percentile of the cumulative distribution

An observation window of $T = 36$ with rolling 12-month windows corresponds to roughly three independent (i.e., non-overlapping) observations. The peak of the probability distribution shifts to the right, and the probability of observing an RAD near zero practically vanishes. This makes intuitive sense, since an RAD of zero would require bias statistics of 1 during *every* 12-month sub-period.

Increasing the observation window to 150 months leads to more than 12 completely independent observations. In this case, the distribution becomes narrower, and the peak shifts further to the right. As T increases further still, the central limit theorem assures that the distribution becomes more sharply peaked about the mean value 0.17, with an approximately normal distribution.

In Fig. 15.7, we indicate by vertical lines the critical mean RAD values, denoted $\tilde{Q}_T^C$, for rolling 12-month windows. These represent the 95th percentile of the cumulative RAD distributions. In other words, assuming perfect risk forecasts and normally distributed returns, over a 36-month observation window we would expect to find an RAD above 0.30 only 5% of the time. If we observe an RAD above this value, we reject the null hypothesis that the risk forecast was accurate.

In reality, of course, returns are not normally distributed. In Fig. 15.8, we plot the critical RAD versus size of observation window T , for various levels of kurtosis. Not surprisingly, the normal distribution ($k = 3$) has the smallest critical values. One useful case to consider is the $T \rightarrow \infty$ limit, which converges to a critical value of 0.17 for kurtosis $k = 3$. As kurtosis increases, however, the critical values also increase. For instance, using the same 36-month observation window as before but now assuming a kurtosis $k = 5$, RAD values in excess of 0.38 would occur about 5% of the time.

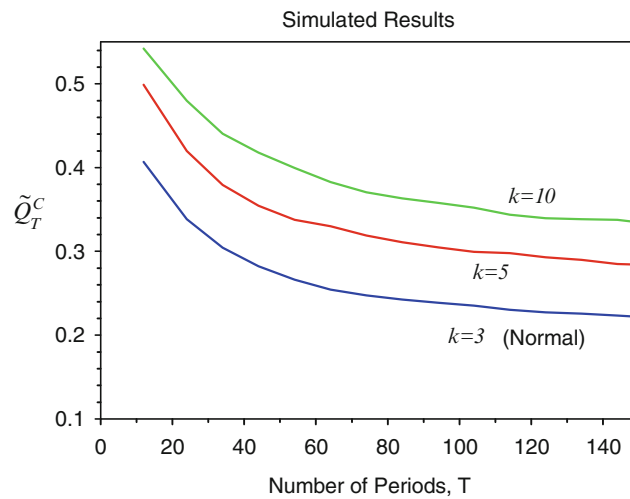


Fig. 15.8 Critical values of rolling absolute deviation (RAD), for rolling 12-month windows, versus the number of periods in the observation window. For perfect risk forecasts and normally distributed returns, over a 60-period observation window we expect to observe an RAD above 0.26 only 5% of the time. For a kurtosis of 10, however, the critical value at 60 periods rises to about 0.40

References

- Dempster, A.P., Laird, N.M., and D.B. Rubin, 1977. "Maximum Likelihood from Incomplete Data via the EM Algorithm," *Journal of the Royal Statistical Society*, vol. 39, no. 1: 1–38.
- Grinold, R., Rudd, A., and D. Stefek, Fall 1989. "Global Factors: Fact or Fiction," *Journal of Portfolio Management*, vol. 16, no. 1: 79–88.
- Heston, S.L., and K.G. Rouwenhorst, 1994. "Does Industrial Structure Explain the Benefits of International Diversification?" *Journal of Financial Economics*, vol. 36: 3–27.
- Markowitz, H.M., March 1952. "Portfolio Selection," *Journal of Finance*, vol. 7, no. 1: 77–91.
- MSCI Barra Fundamental Data Methodology Handbook, 2008, <http://www.msci.com/methodology/meth.docs/MSCI.Sep08.Fundamental.Data.pdf>
- Newey, W., and K. West, 1987. "A Simple, Positive Semi-Definite, Heteroskedasticity and Auto-correlation Consistent Covariance Matrix," *Econometrica*, vol. 55, no. 3: 703–708.
- Rosenberg, B., March 1974. "Extra-Market Components of Covariance Among Security Prices," *Journal of Financial and Quantitative Analysis*, vol. 9: 263–274.
- Rosenberg, B., and V. Marathe, May 1975. "Tests of Capital Asset Pricing Hypotheses," *Research Program in Finance*, Working Paper no. 32.
- Rudd, A., and B. Rosenberg, 1979. "Realistic Portfolio Optimization," in Elton, E., and M. Gruber, (editors): *Portfolio Theory, 25 Years After, Studies in the Management Sciences*, vol. 11, North Holland, Amsterdam: 21–46.
- Sharpe, W.F., September 1964. "Capital Asset Prices: A Theory of Market Equilibrium Under Conditions of Risk," *Journal of Finance*, vol. 19, no. 3: 425–442.
- Tobin, J., February 1958. "Liquidity Preference as a Behavior Toward Risk," *Review of Economic Studies*, vol. 67: 65–86.